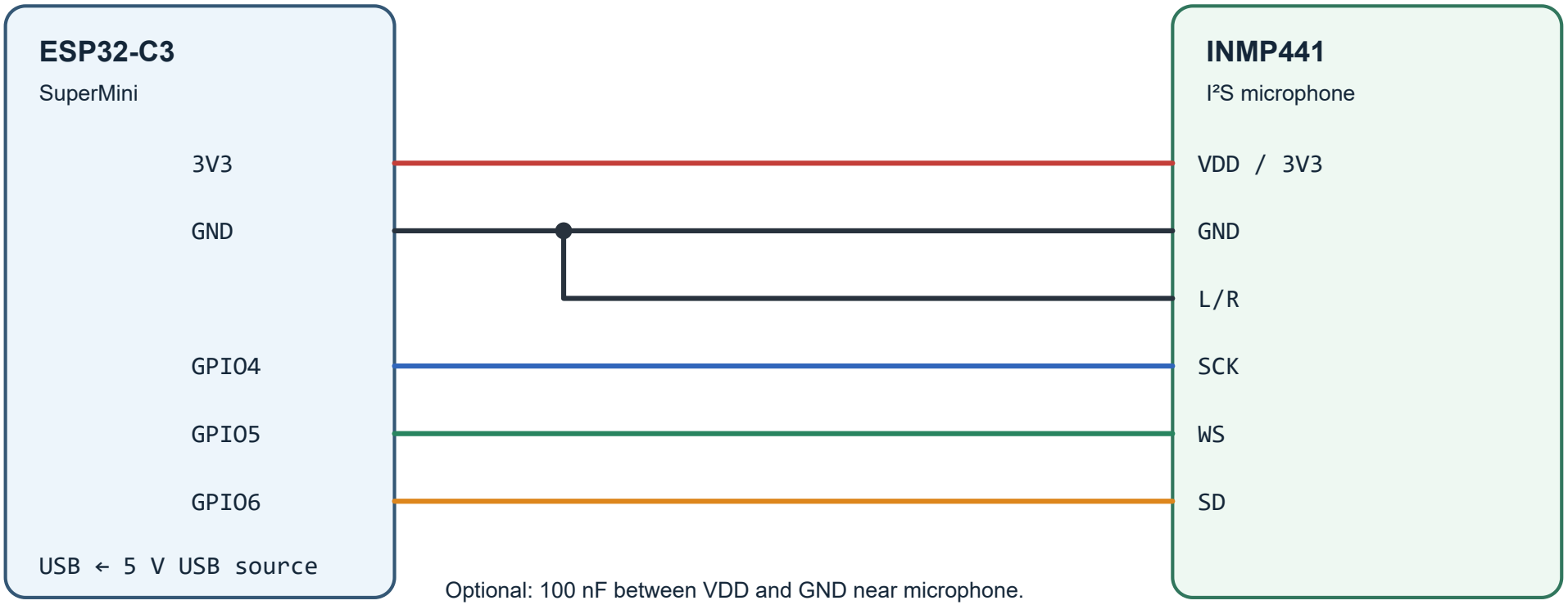
1 / USB test: one cot node

ESP32-C3 SuperMini ↔ INMP441 microphone

Repeat this same signal wiring for each node. Only one node has been tested end-to-end. Labels show electrical pins, **not physical header positions**.



Recorded working wiring: L/R is tied to ground (left channel). The old document's GPIO5/6/7 mapping is wrong for current firmware.

Switch power off before wiring. Match the labels on the actual modules. Keep microphone opening unobstructed. Wire colours are illustrative.

2 / USB test: central hub

ESP32-S3-DevKitC-1 ↔ existing 2.8-inch ILI9341 SPI display

Existing panel wiring recorded on the project. Pin positions below are symbolic; check the labels and power requirements of your exact display.



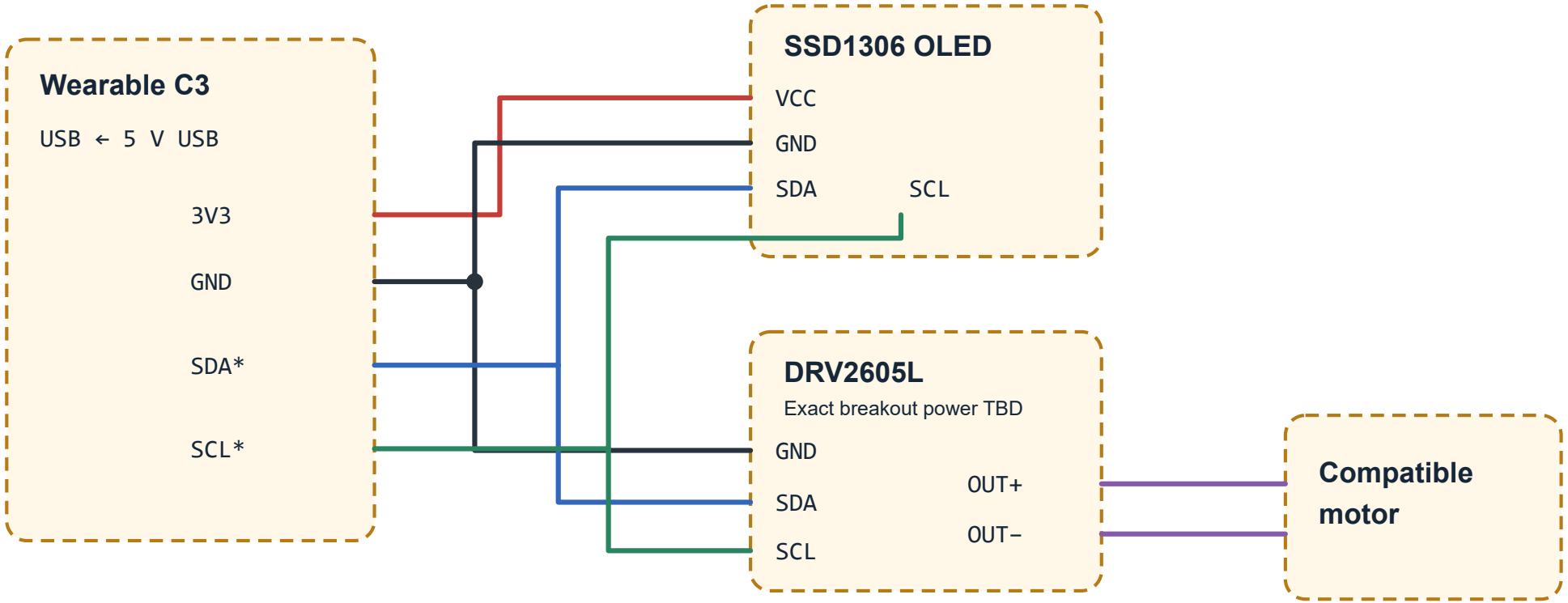
Backlight caution: GPIO4 controlled the existing panel. Do not connect an unknown bare backlight LED to a GPIO; verify the module's current and circuitry. The older 100 Ω resistor and 5 V VCC are not universal.

The hub and node communicate by ESP-NOW; no signal wire or common ground is required between them.

3 / USB test: wearable concept

ESP32-C3 + SSD1306 OLED + DRV2605L + compatible motor

PROPOSED, NOT YET BUILT OR TESTED: GPIO numbers and breakout power pins are not selected. This is a bus diagram, not a finished pin map.



* Choose GPIOs after checking board, boot and USB constraints.

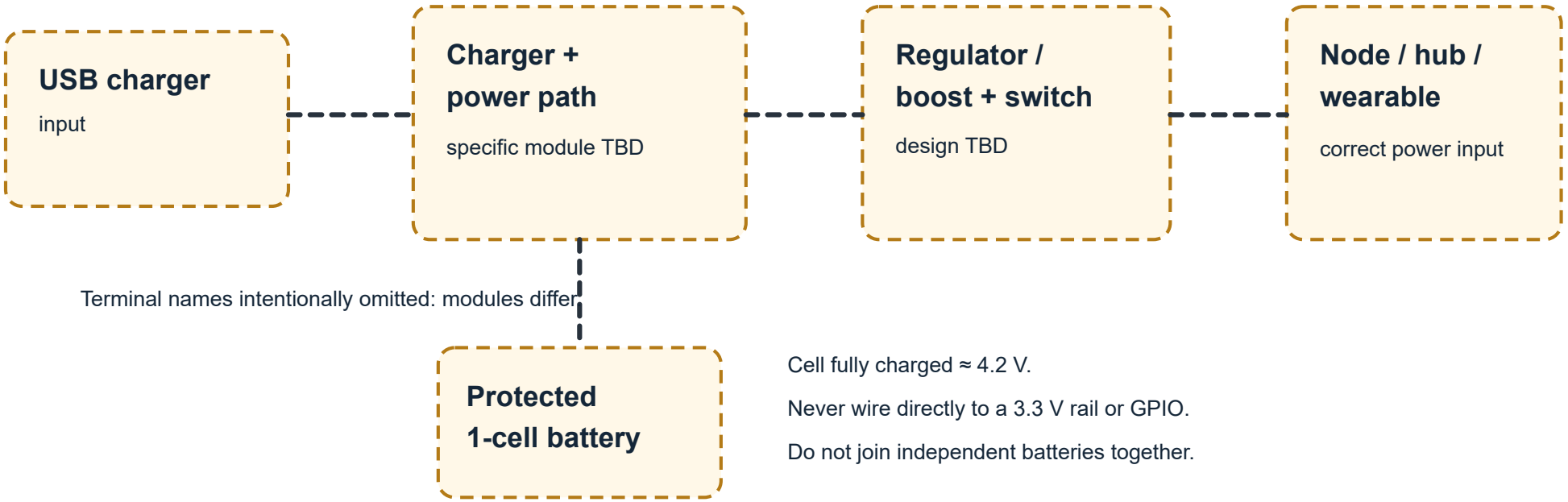
The driver needs the correct supply and 3.3 V-safe I²C pull-ups for its **specific breakout**; verify EN and other pins from that board's documentation. Connect motor across driver outputs—not to ground as in the old MOSFET design.

Dashed outlines = unverified design. The wearable firmware has not been written. Do not treat this page as a final assembly schematic.

4 / Proposed battery-powered system

Power architecture for each unit — not an exact charger wiring schematic

Each cot, hub and wearable has its own battery and local supply. Retain the verified signal wiring from pages 1–2.



Do not wire a lithium charger from this block diagram. Select and verify each real module, cell, charge current, load sharing, regulator, switch and USB backfeed behaviour first. Basic TP4056 protection boards are not automatically power-path circuits.

Battery monitoring and status LED GPIOs are unassigned; current firmware does not implement them. See SYSTEM_WIRING.md for the candidate divider and open decisions.